

Analog Communications

MATLAB Simulation Report

DSB, SSB, and NBFM: Generation, Detection, and Noise Performance

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1. Abstract:

This report presents a comprehensive MATLAB implementation and analysis of three analog modulation schemes: Double Sideband (DSB), Single Sideband (SSB), and Narrowband Frequency Modulation (NBFM). The project investigates generation, demodulation, and performance analysis under various noise conditions, frequency errors, and phase errors. Key findings include the superiority of coherent detection over envelope detection for DSB-SC, the bandwidth efficiency of SSB, and the noise immunity of NBFM. All implementations follow the assignment specifications with carrier frequency $F_c = 100\text{ kHz}$ and sampling frequency $F_s = 5 \times F_c = 500\text{ kHz}$.

2. Introduction:

Analog modulation is fundamental to communication systems, enabling the transmission of baseband signals over bandpass channels. This project explores three key modulation techniques:

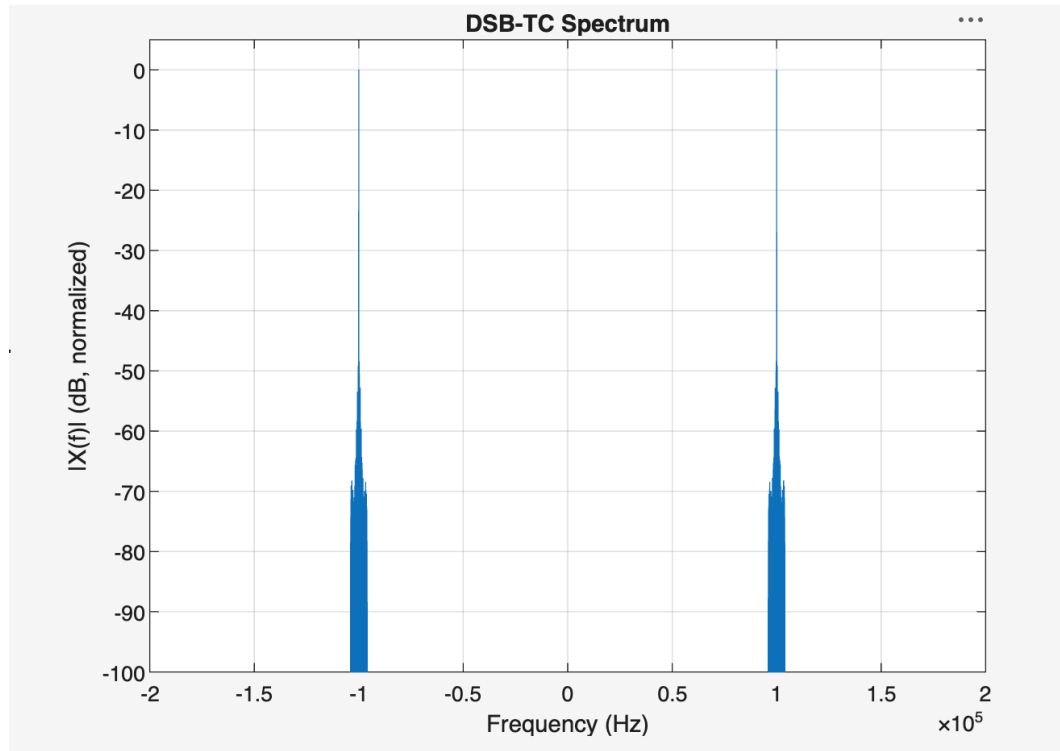
Modulation	Description	Key Feature
DSB-TC	Double Sideband Transmitted Carrier	Simple envelope detection
DSB-SC	Double Sideband Suppressed Carrier	Power efficient, requires coherent detection
SSB	Single Sideband	Bandwidth efficient (half of DSB)
NBFM	Narrowband Frequency Modulation	Good noise immunity

The experiments use an audio file sampled at 48 kHz, band limited to 4 kHz, and modulated onto a 100 kHz carrier with $F_s = 500\text{ kHz}$.

3. Experiment 1: Double Sideband (DSB) Modulation:

3.1 DSB-TC Generation

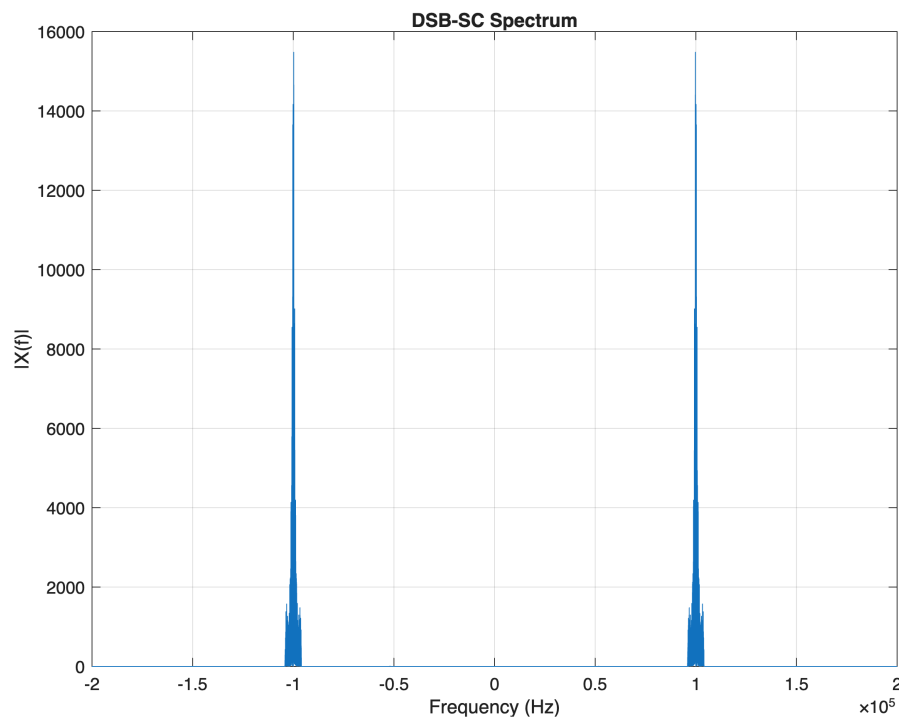
Figure 1: DSB-TC Spectrum



Observation: The DSB-TC spectrum shows the carrier at ± 100 kHz with two sidebands extending from 96-100 kHz and 100-104 kHz. The carrier is clearly visible as a strong peak, confirming the transmitted carrier component.

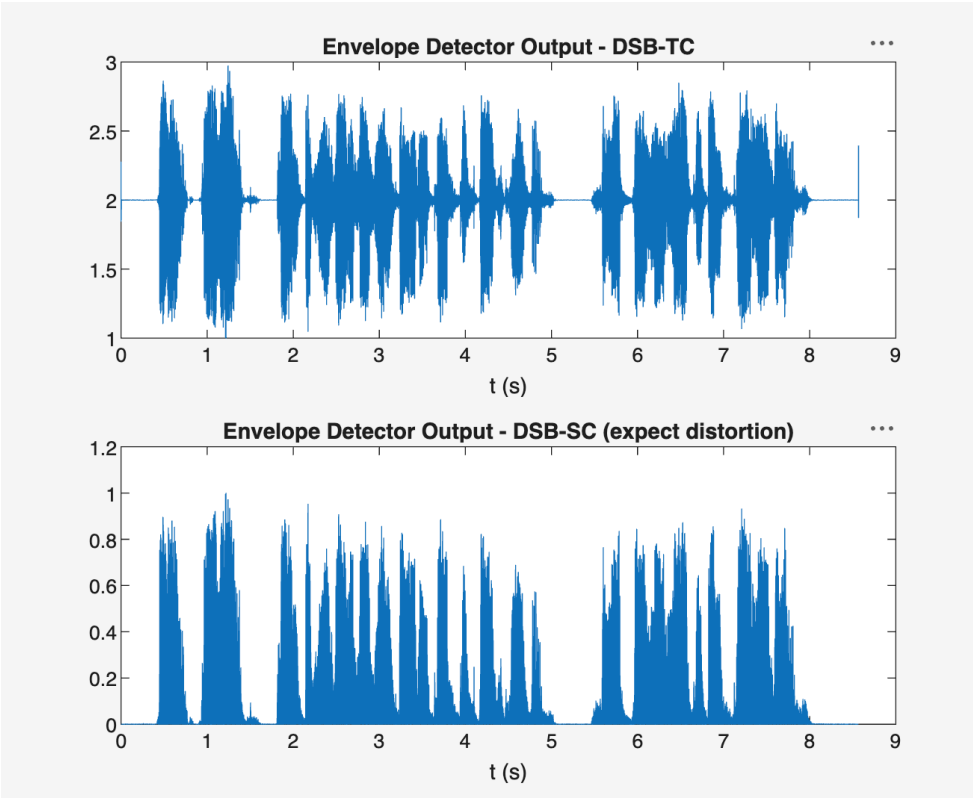
3.1 DSB-SC Generation

Figure 2: DSB-SC Spectrum



Observation: The DSB-SC spectrum shows only the two sidebands at ± 100 kHz. The carrier is suppressed, resulting in a 3 dB power saving compared to DSB-TC but requiring more complex demodulation.

Figure 3: Envelope Detector Outputs (DSB-TC vs DSB-SC)



Modulation Type	Envelope Detection Result
DSB-TC	Clean recovery of message
DSB-SC	Distorted (rectified) output

Observation: Envelope detection works perfectly for DSB-TC because the carrier provides a DC bias that keeps the signal envelope positive. For DSB-SC, the signal crosses zero, causing the envelope detector to produce a rectified version with severe distortion. This demonstrates that envelope detection is only suitable for carrier-transmitted schemes (DSB-TC), not suppressed-carrier schemes (DSB-SC).

3.3 Coherent Detection with SNR

Figure 4: Coherent Detection - SNR = 0 dB

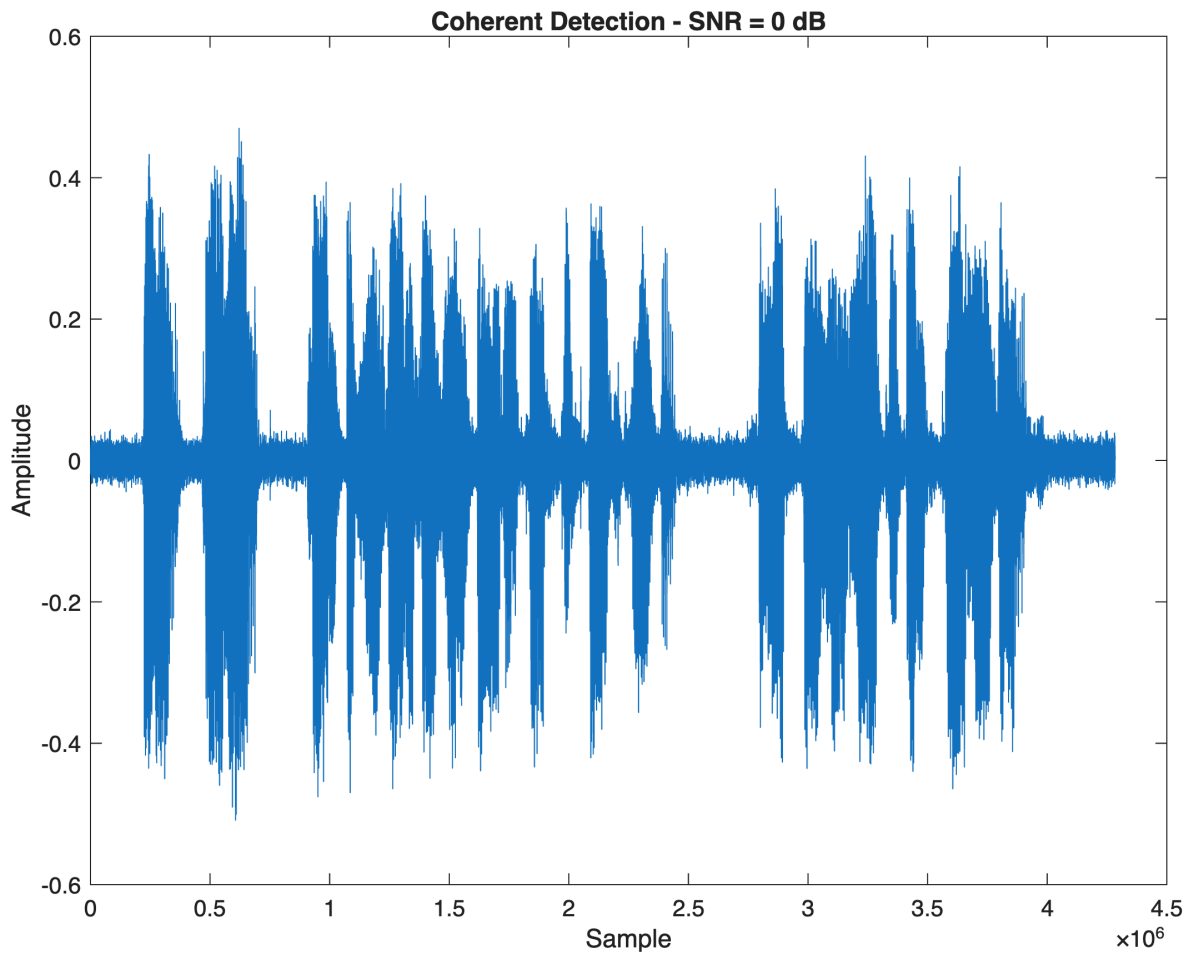
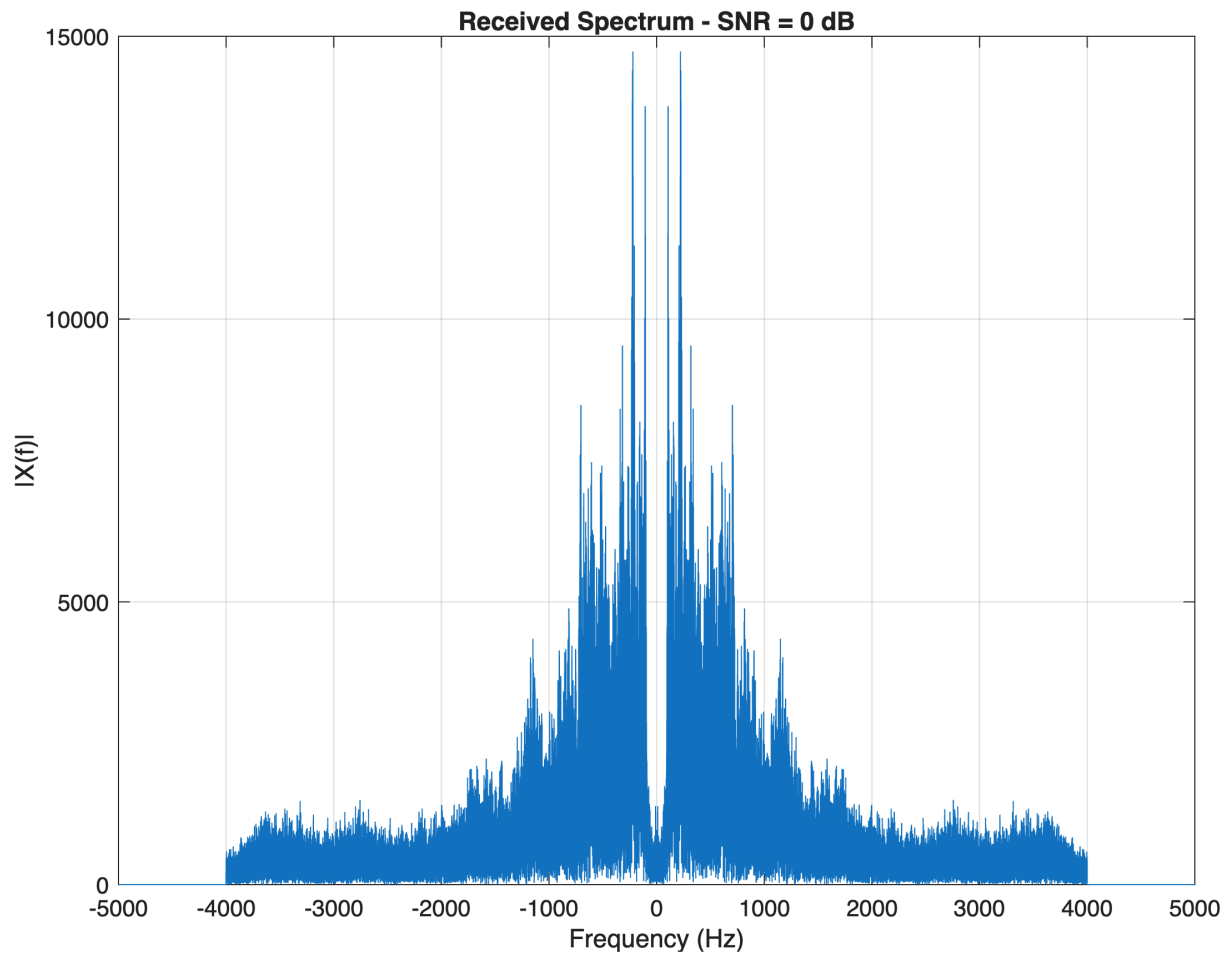


Figure 5: Received Spectrum - SNR = 0 dB

Observation: At SNR = 0 dB, the signal is completely buried in noise. The time-domain signal shows random fluctuations with no discernible pattern, and the spectrum shows a high noise floor masking the message.

Figure 6: Coherent Detection - SNR = 10 dB

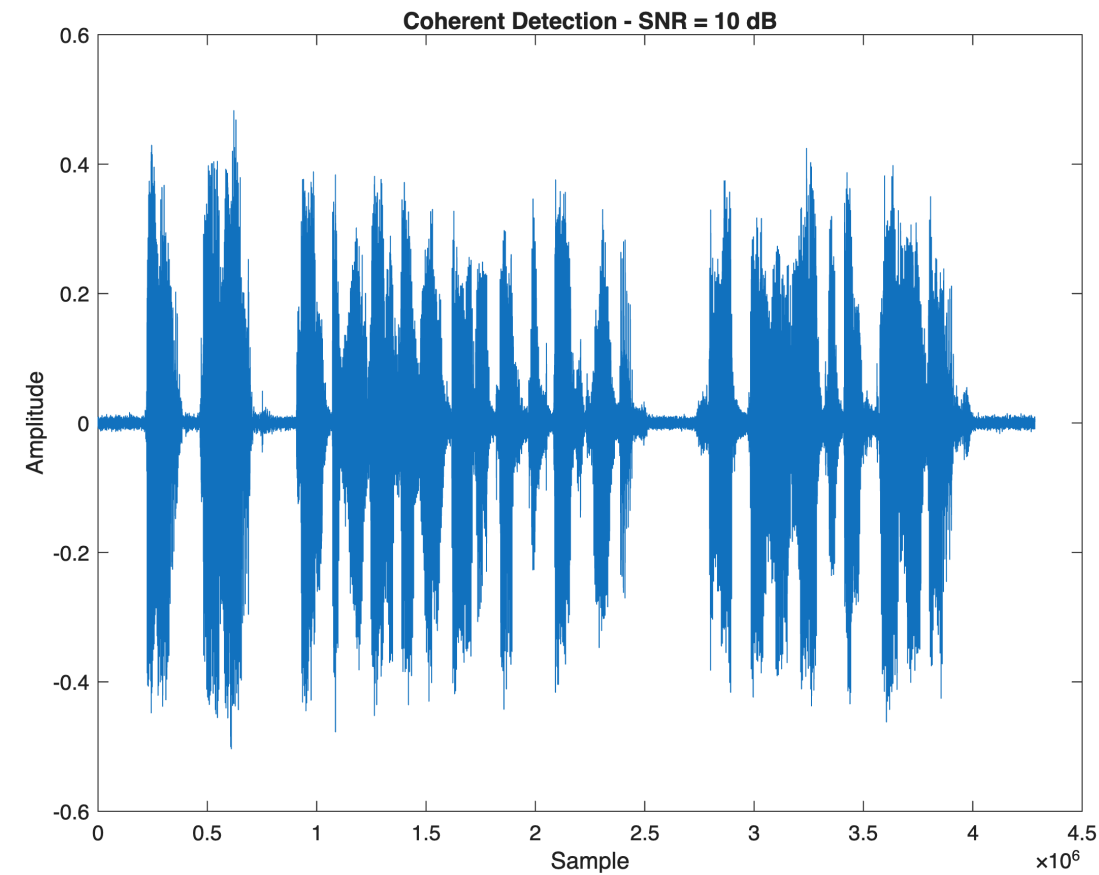
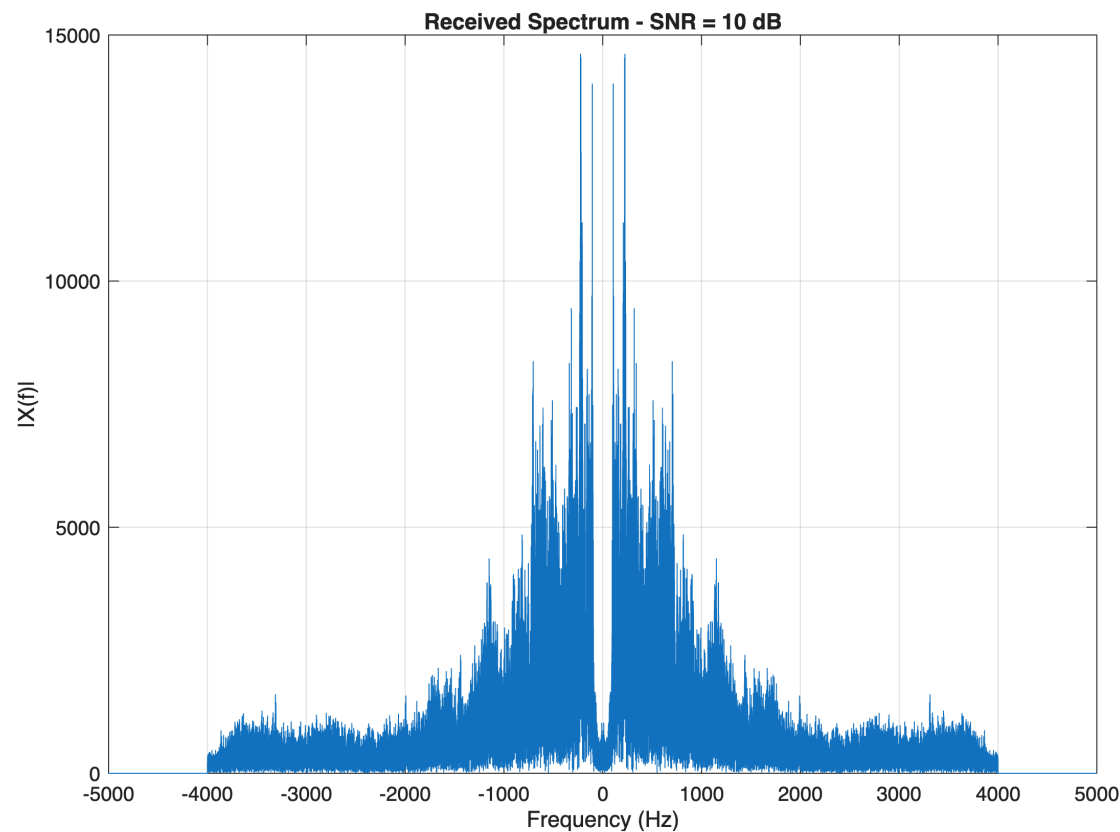


Figure 7: Received Spectrum - SNR = 10 dB



Observation: At SNR = 10 dB, the message becomes visible but with significant noise. The spectrum shows peaks at the message frequencies with a noticeable noise floor.

Figure 8: Coherent Detection - SNR = 30 dB

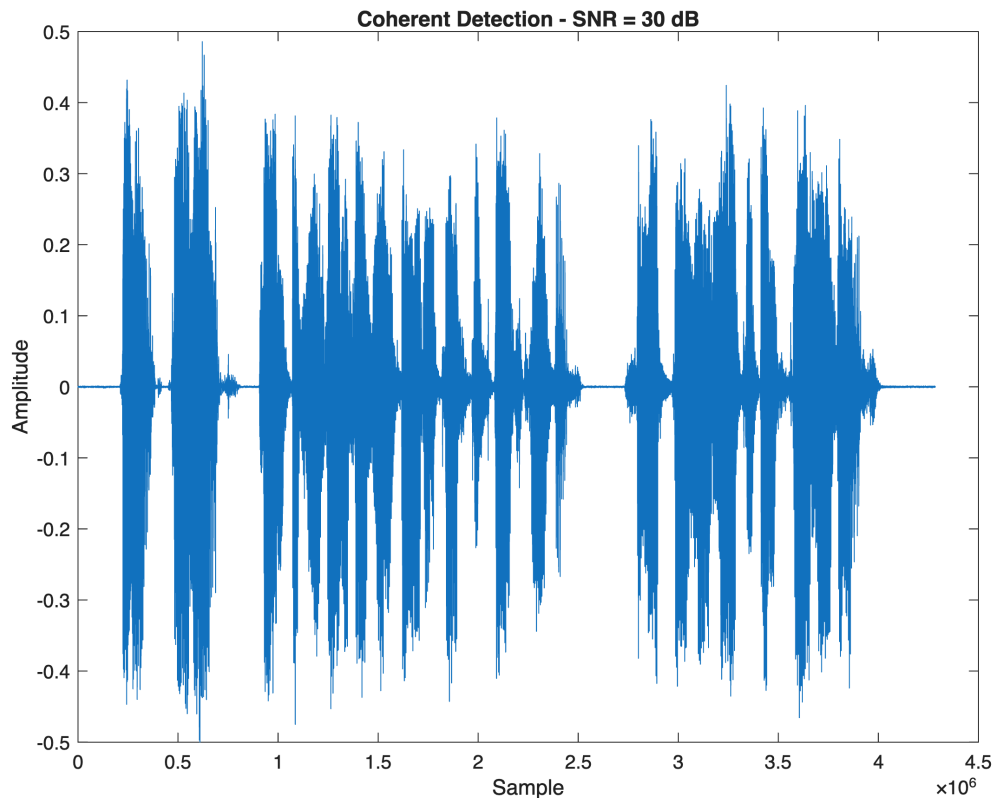
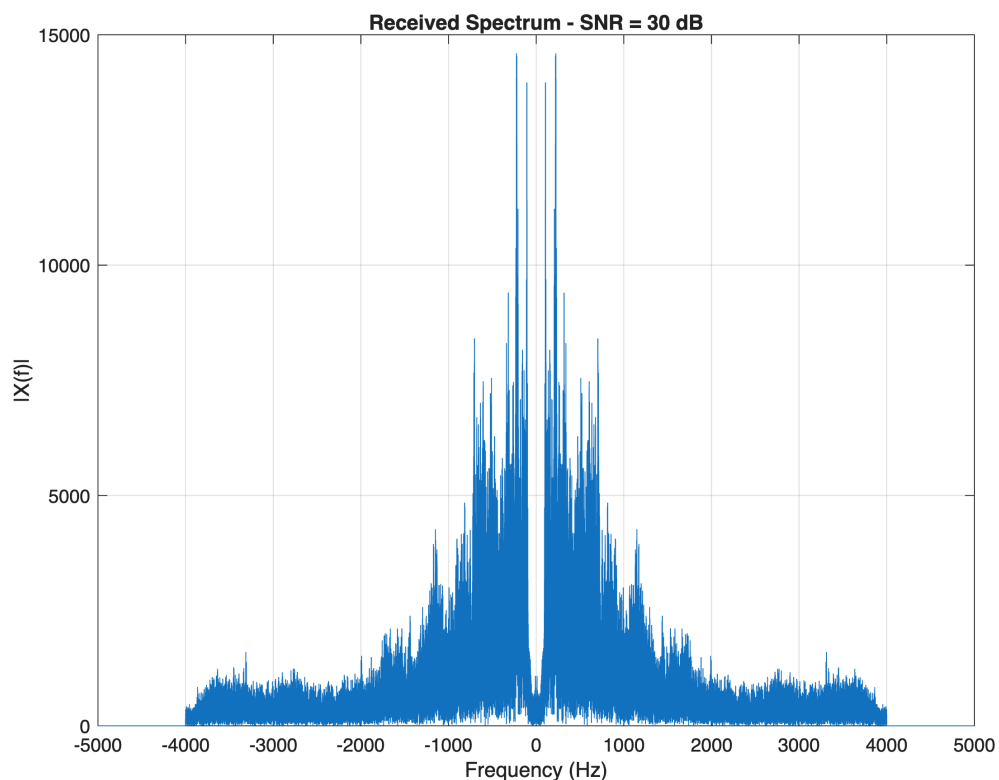


Figure 9: Received Spectrum - SNR = 30 dB



Observation: At SNR = 30 dB, the recovered signal is clean and nearly identical to the original. The spectrum shows clear peaks corresponding to the message frequencies with a very low noise floor.

SNR Performance Summary

SNR (dB)	Audio Quality	Spectrum Quality
0 dB	Very noisy, voice barely audible	Signal buried in noise
10 dB	Audible with noticeable static	Peaks visible with noise
30 dB	Clean voice	Clear peaks, low noise floor

Conclusion: Coherent detection performs well under high SNR conditions. The signal quality improves significantly as SNR increases, demonstrating the importance of good signal-to-noise ratio in communication systems.

3.4 Frequency Error (100.1 kHz):

Figure 10: Coherent Detection - Frequency Error (100.1 kHz LO)

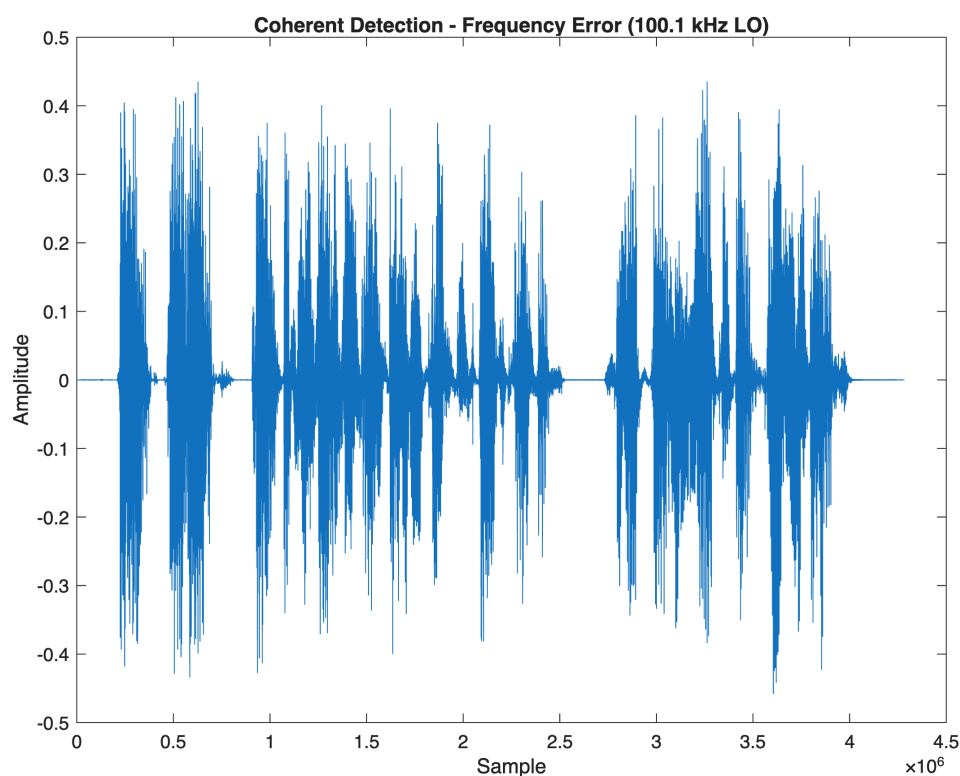
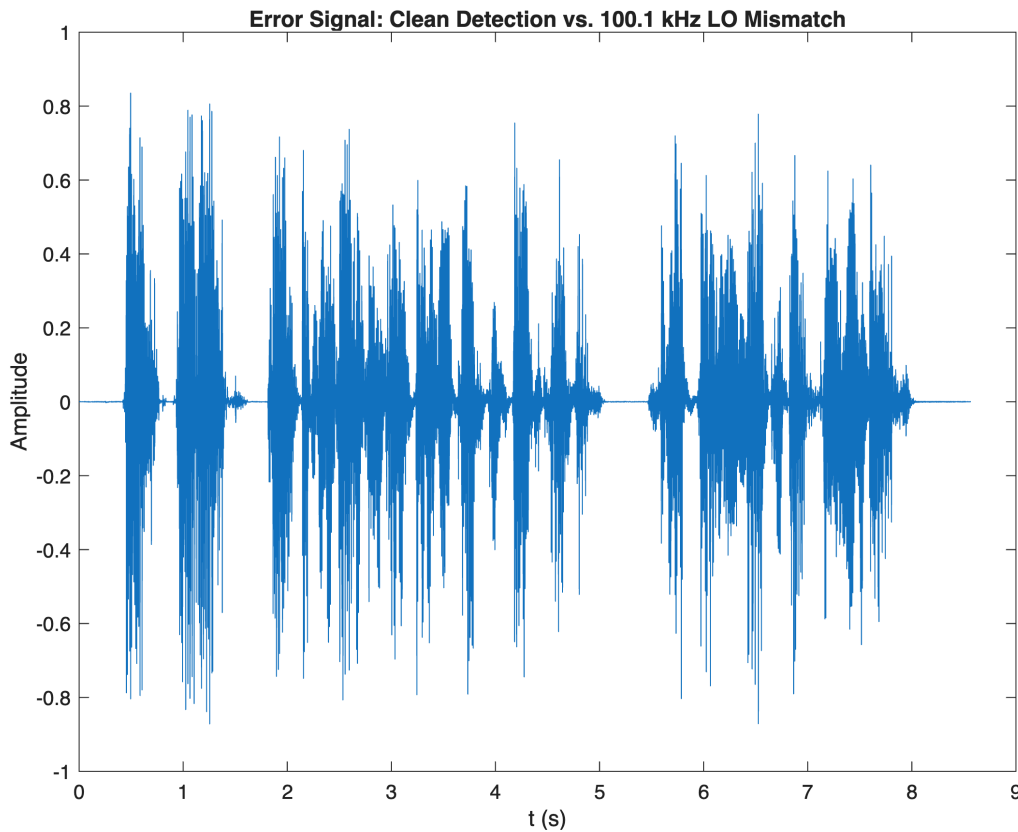


Figure 11: Error Signal: Clean Detection vs. 100.1 kHz LO Mismatch

Observation: When the local oscillator has a 100 Hz frequency error (100.1 kHz instead of 100 kHz), the recovered signal shows a beat frequency pattern. The message envelope is modulated at 100 Hz, which corresponds to the frequency difference between the carrier and the LO.

$$\cos(2\pi \times 100\text{k} \times t) \times \cos(2\pi \times 100.1\text{k} \times t) = 0.5[\cos(2\pi \times 200.1\text{k} \times t) + \cos(2\pi \times 100 \times t)]$$

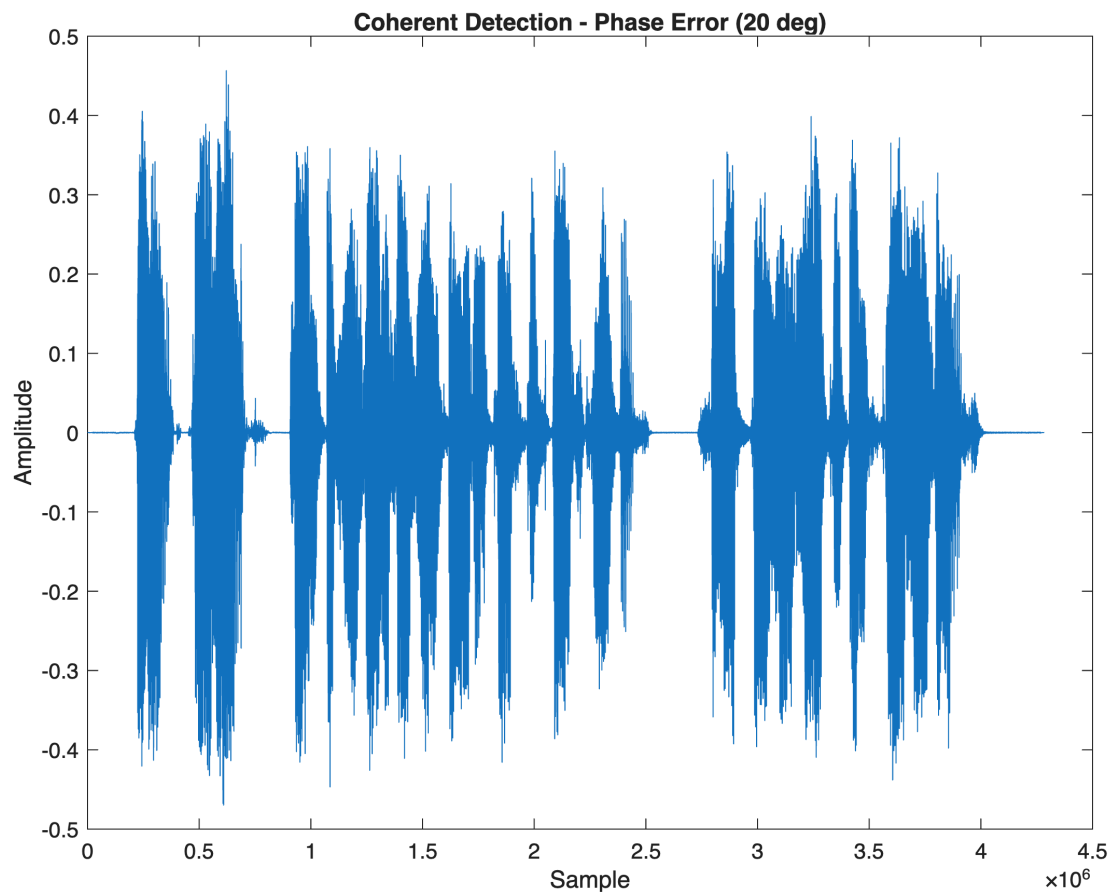
After low-pass filtering, the 200.1 kHz component is removed, leaving the 100 Hz beat frequency. This phenomenon is known as the heterodyne effect or beat frequency phenomenon.

RMS error from 100 Hz frequency offset: 0.1094

Conclusion: Frequency synchronization is critical in coherent detection. Even a small 100 Hz offset creates a 100 Hz beat frequency that distorts the recovered message.

3.5 Phase Error (20°)

Figure 12: Coherent Detection - Phase Error (20 deg)



Observation: A 20° phase error causes amplitude attenuation of the recovered signal but does NOT introduce a beat frequency.
The attenuation factor is: $\cos(20^\circ) = 0.94$

Comparison: Frequency Error vs. Phase Error

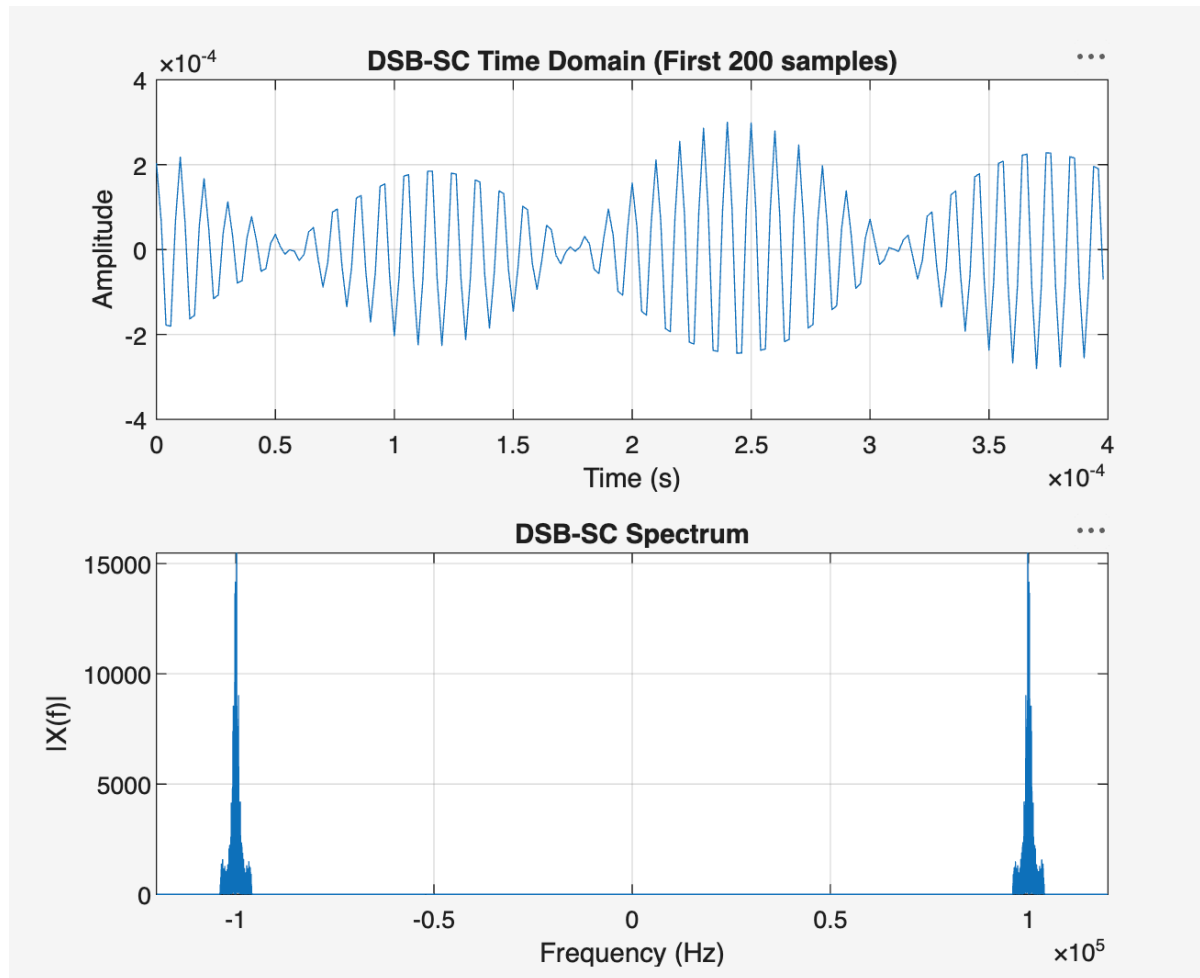
Error Type	Effect on Recovered Signal
Frequency Error (100 Hz)	Beat frequency at 100 Hz
Phase Error (20°)	Amplitude reduction ($\cos 20^\circ = 0.94$)

Conclusion: Frequency errors are more severe than phase errors because they introduce time-varying distortion (beat frequency), whereas phase errors only cause constant amplitude reduction.

4. Experiment 2: Single Sideband (SSB) Modulation:

4.1 DSB-SC Generation

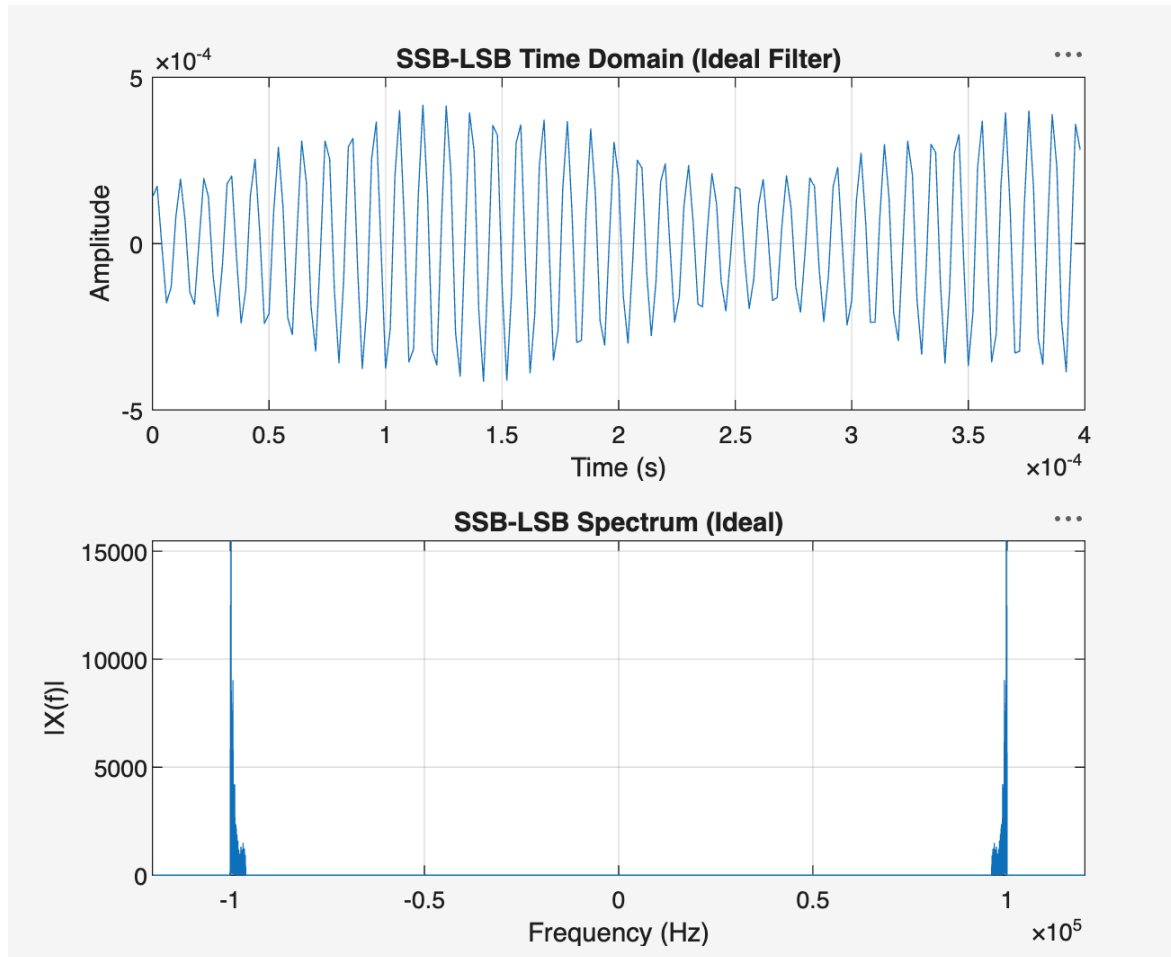
Figure 13: DSB-SC Spectrum



Observation: The DSB-SC spectrum shows two sidebands around ± 100 kHz, each occupying 4 kHz bandwidth (96-100 kHz and 100-104 kHz). Total bandwidth = 8 kHz.

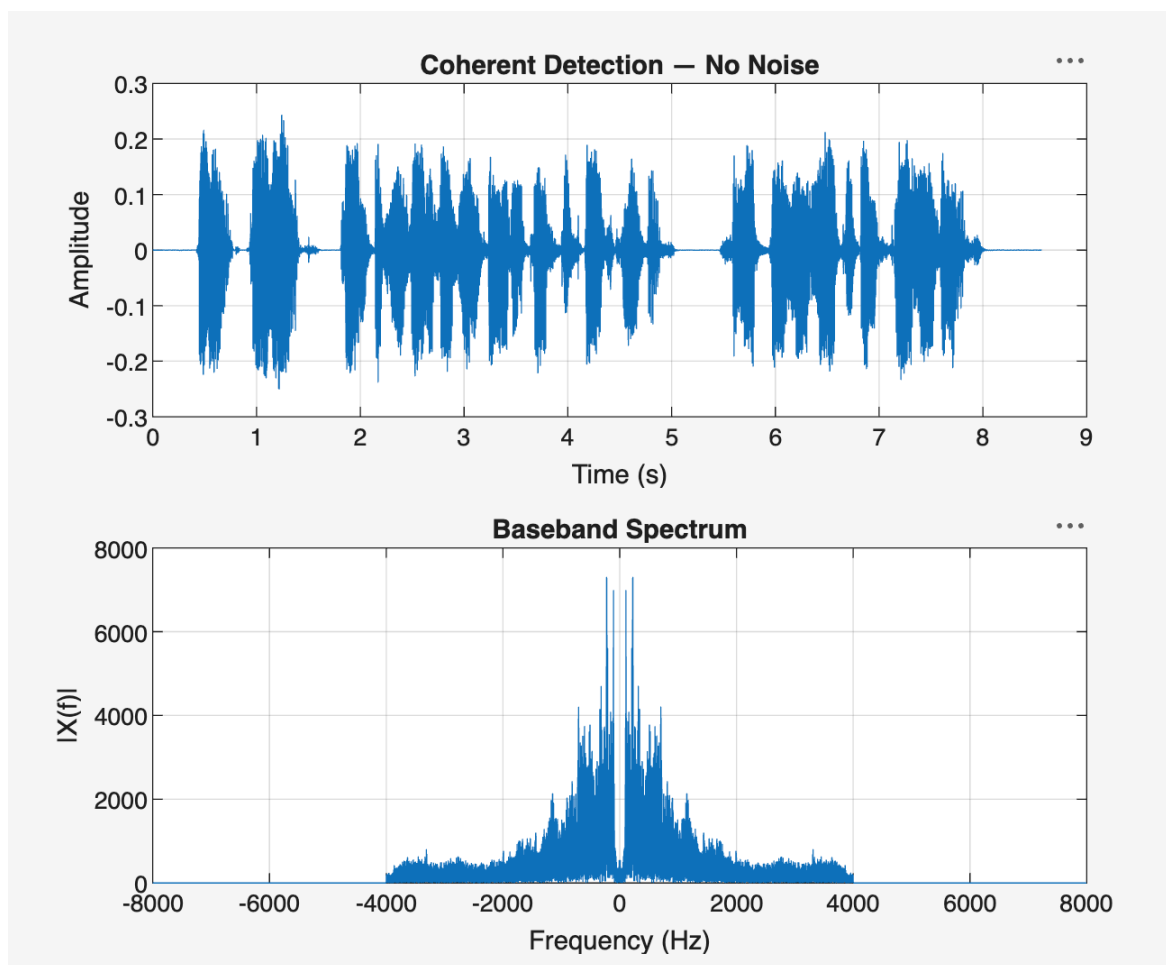
4.2 Ideal SSB Filtering

Figure 14: SSB-LSB (Ideal Filter)



Observation: The ideal filter completely removes the upper sideband (USB), leaving only the lower sideband (LSB). The total bandwidth is now only 4 kHz (96-100 kHz), achieving 50% bandwidth reduction compared to DSB.

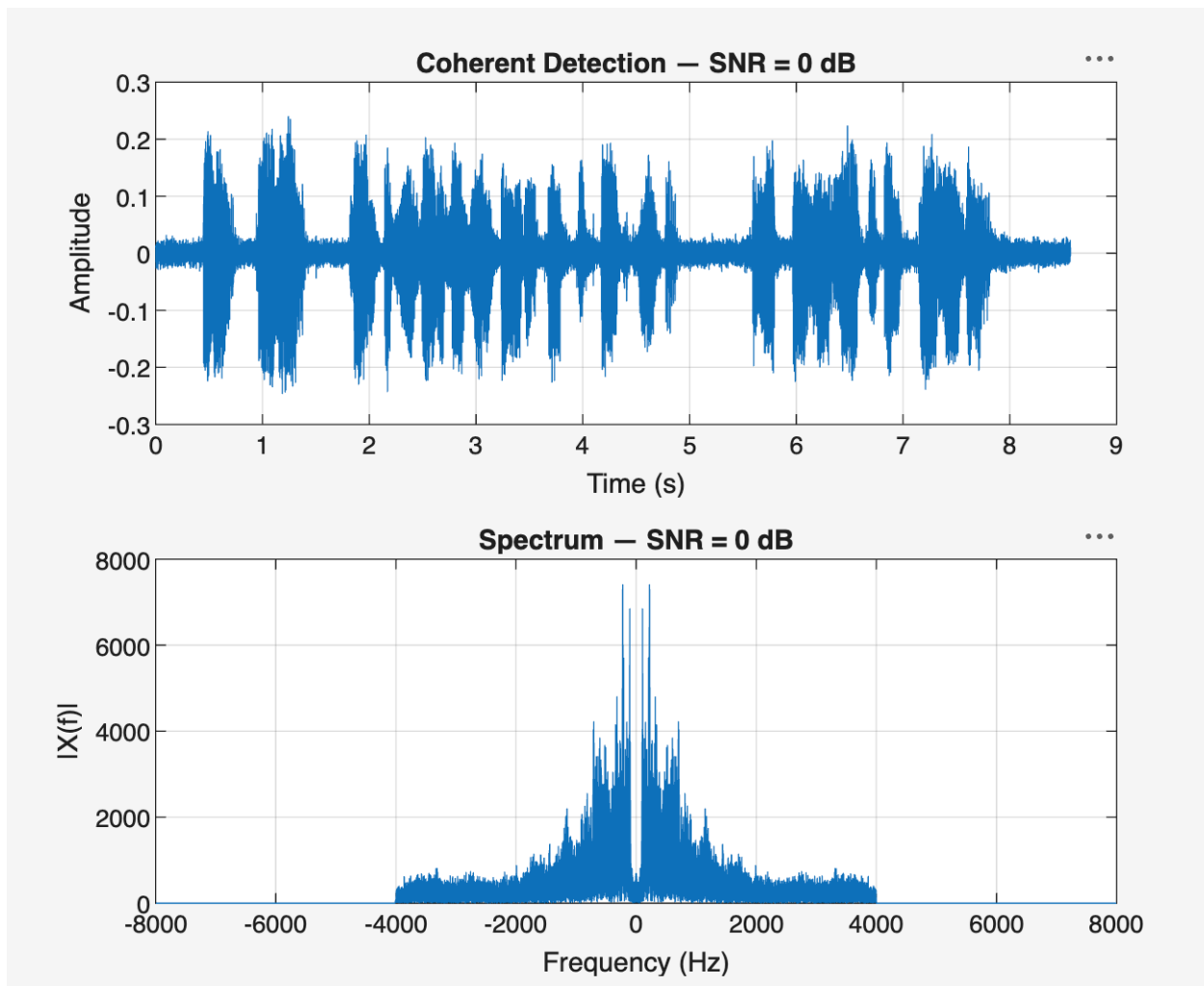
The time-domain signal shows the single-sideband structure with no carrier component.

Figure 15: Coherent Detection - No Noise

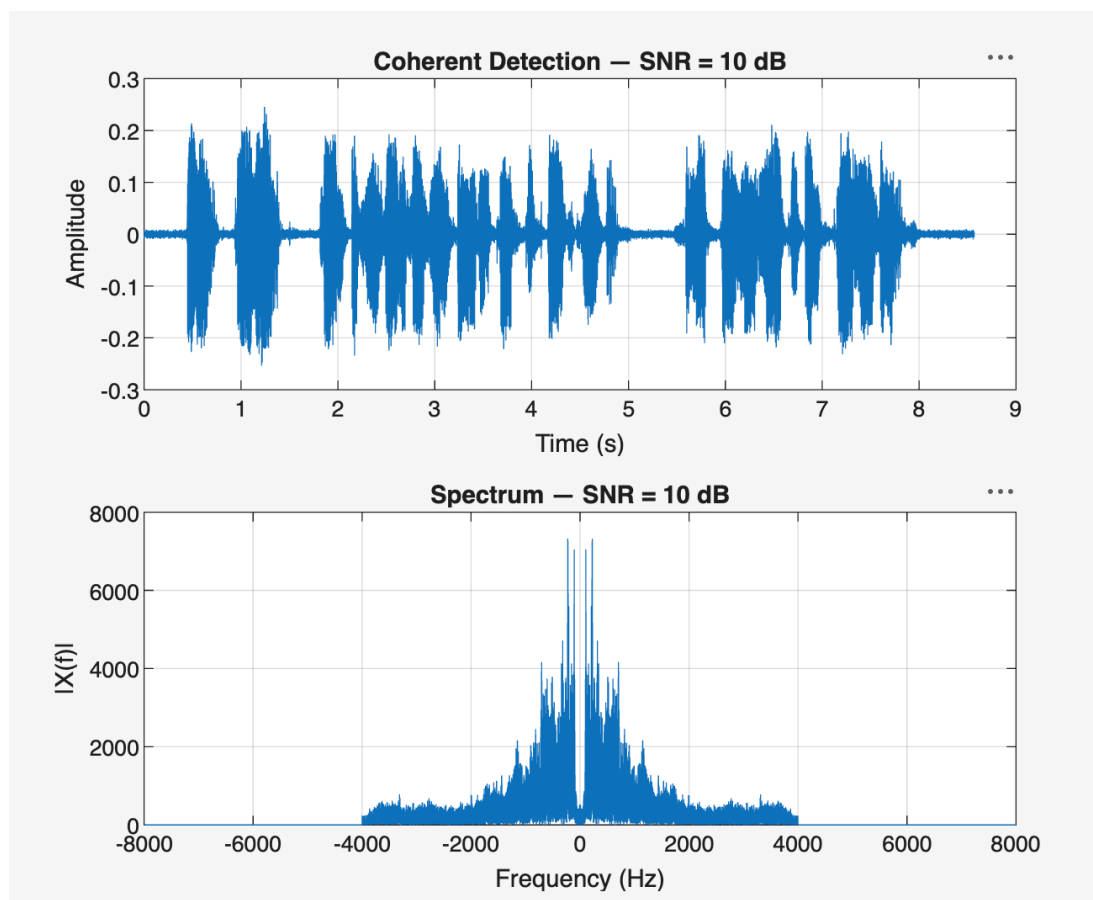
Observation: With no noise, coherent detection perfectly recovers the original baseband signal. The spectrum shows the recovered 4 kHz bandwidth centered at 0 Hz.

4.4 Coherent Detection with SNR

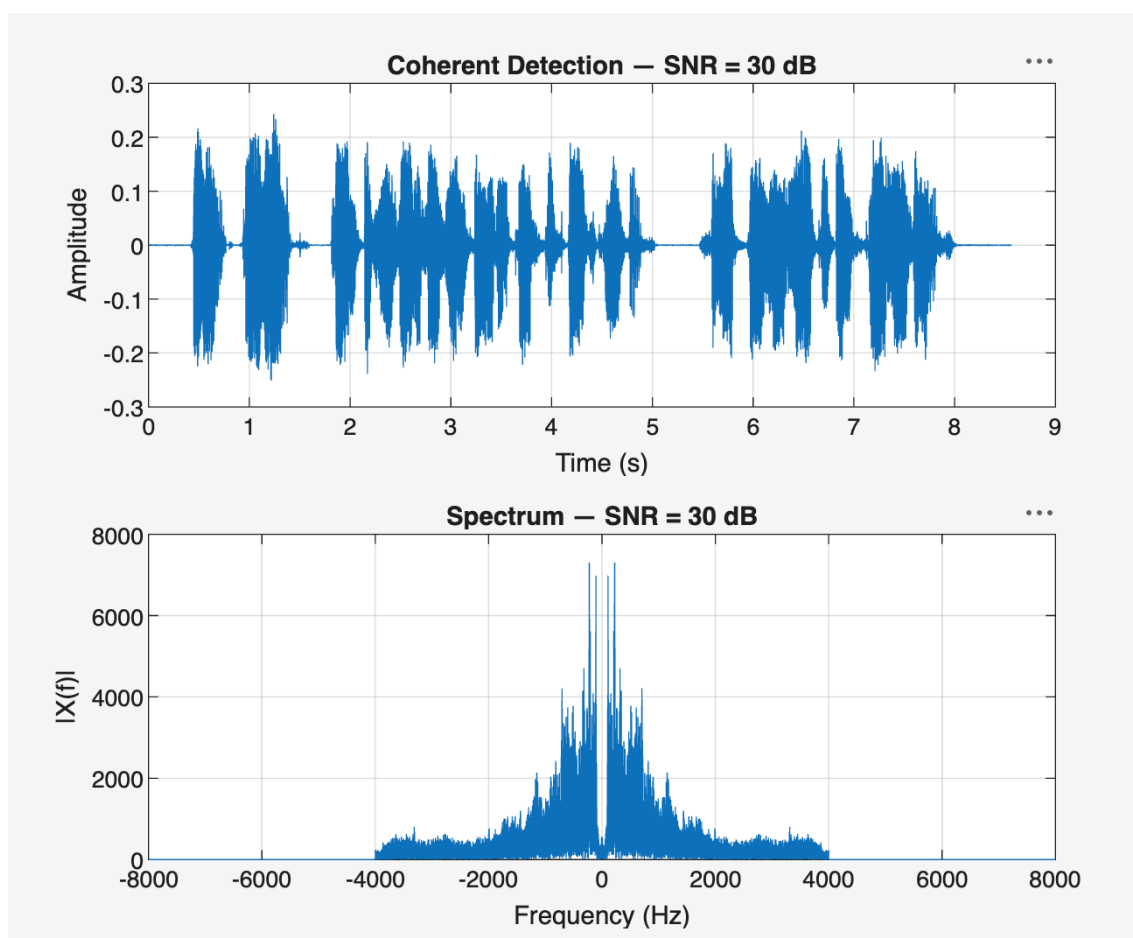
Figure 16: Coherent Detection - SNR = 0 dB



Observation: At 0 dB SNR, the signal is overwhelmed by noise. The voice is barely audible

Figure 17: Coherent Detection - SNR = 10 dB

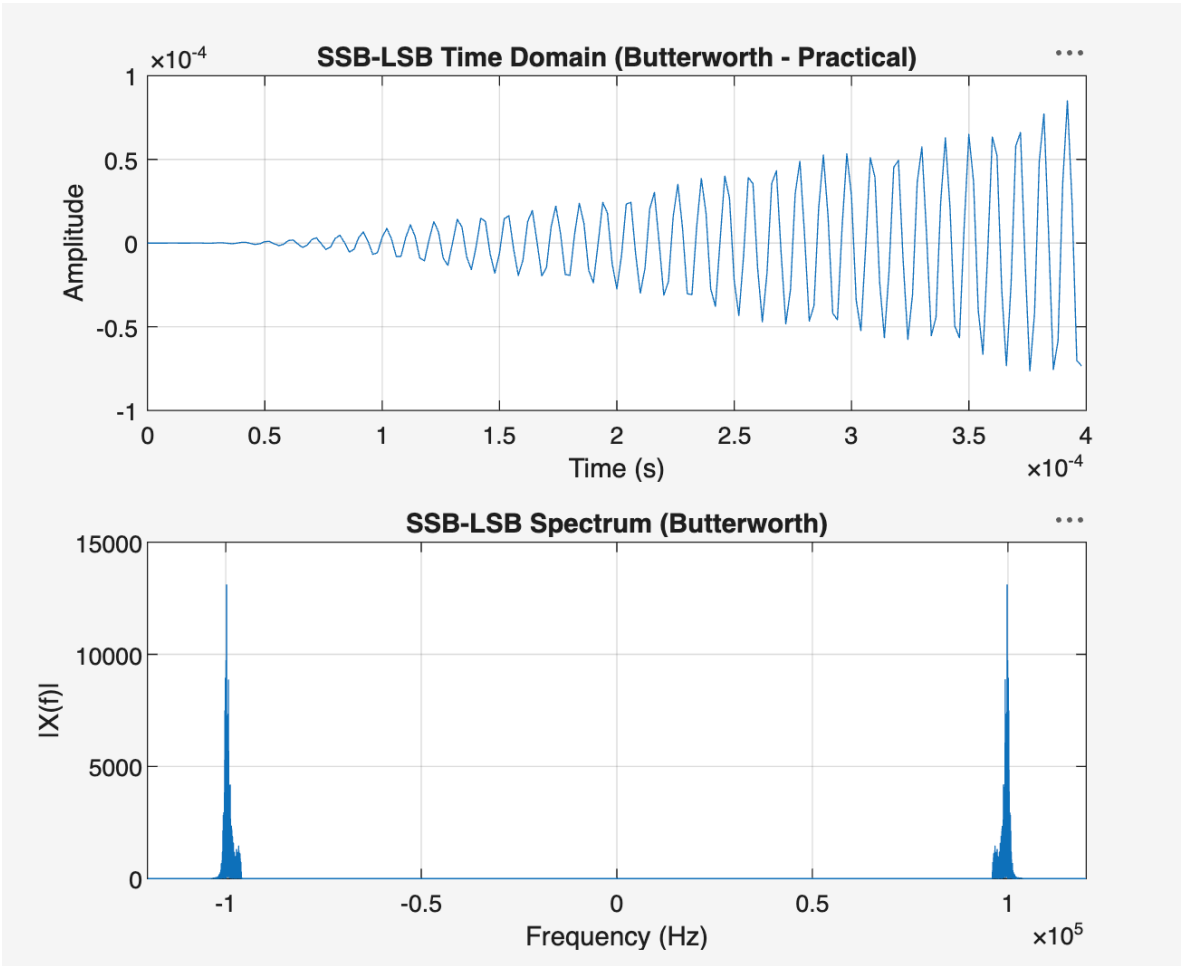
Observation: At 10 dB SNR, the signal is audible but with noticeable background noise. The spectrum shows signal peaks above the noise floor.

Figure 18: Coherent Detection - SNR = 30 dB

Observation: At 30 dB SNR, the recovered signal is clear and clean. The spectrum shows sharp peaks with very low noise floor

4.5 Practical Butterworth Filter

Figure 19: SSB-LSB Spectrum (Butterworth)



Comparison: Ideal vs. Butterworth Filter

Aspect	Ideal Filter	Butterworth Filter
Transition Band	None (brick-wall)	Gradual (finite slope)
Phase Distortion	None	Present (non-linear phase)
USB Rejection	Infinite	Finite (attenuation)
Implementation	Non-causal (FFT)	Causal (filter)

Observation: The Butterworth filter (4th order, using `filter()`) shows:

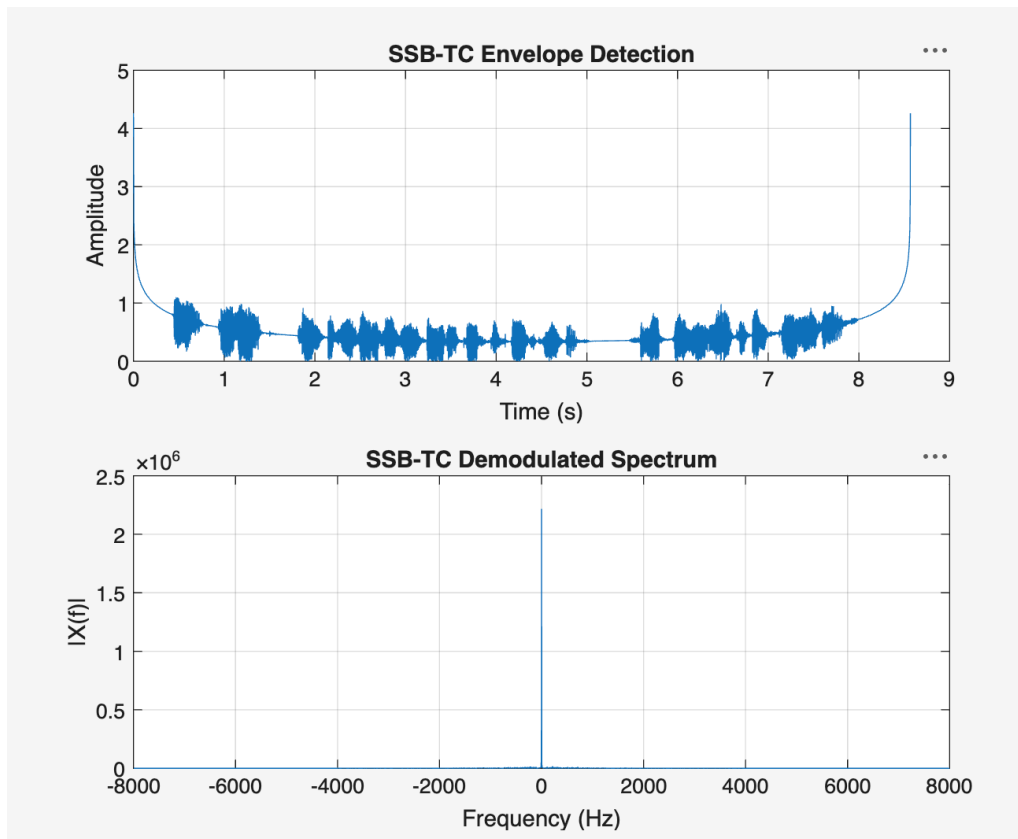
- Residual USB leakage due to finite transition band
- Phase distortion causing slight waveform distortion
- Passband ripple affecting signal quality
- More practical for real-world implementation

Conclusion: While the ideal filter provides perfect USB rejection, it's not realizable in practice. The Butterworth filter demonstrates the trade-offs in practical filter design.

For this implementation, the 4th-order Butterworth filter was configured with cutoff frequencies designed to specifically isolate the 96 kHz to 100 kHz lower sideband (LSB). Utilizing MATLAB's `butter()` and `filter()` functions allowed for the creation of this bandpass response, demonstrating how theoretical, non-causal brick-wall requirements must be adapted into realizable digital signal processing constraints.

4.6 SSB-TC with Envelope Detection

Figure 20: SSB-TC Envelope Detection



Observation: SSB-TC (with transmitted carrier) can be demodulated using a simple envelope detector, similar to DSB-TC. The carrier provides the necessary DC bias for envelope detection.

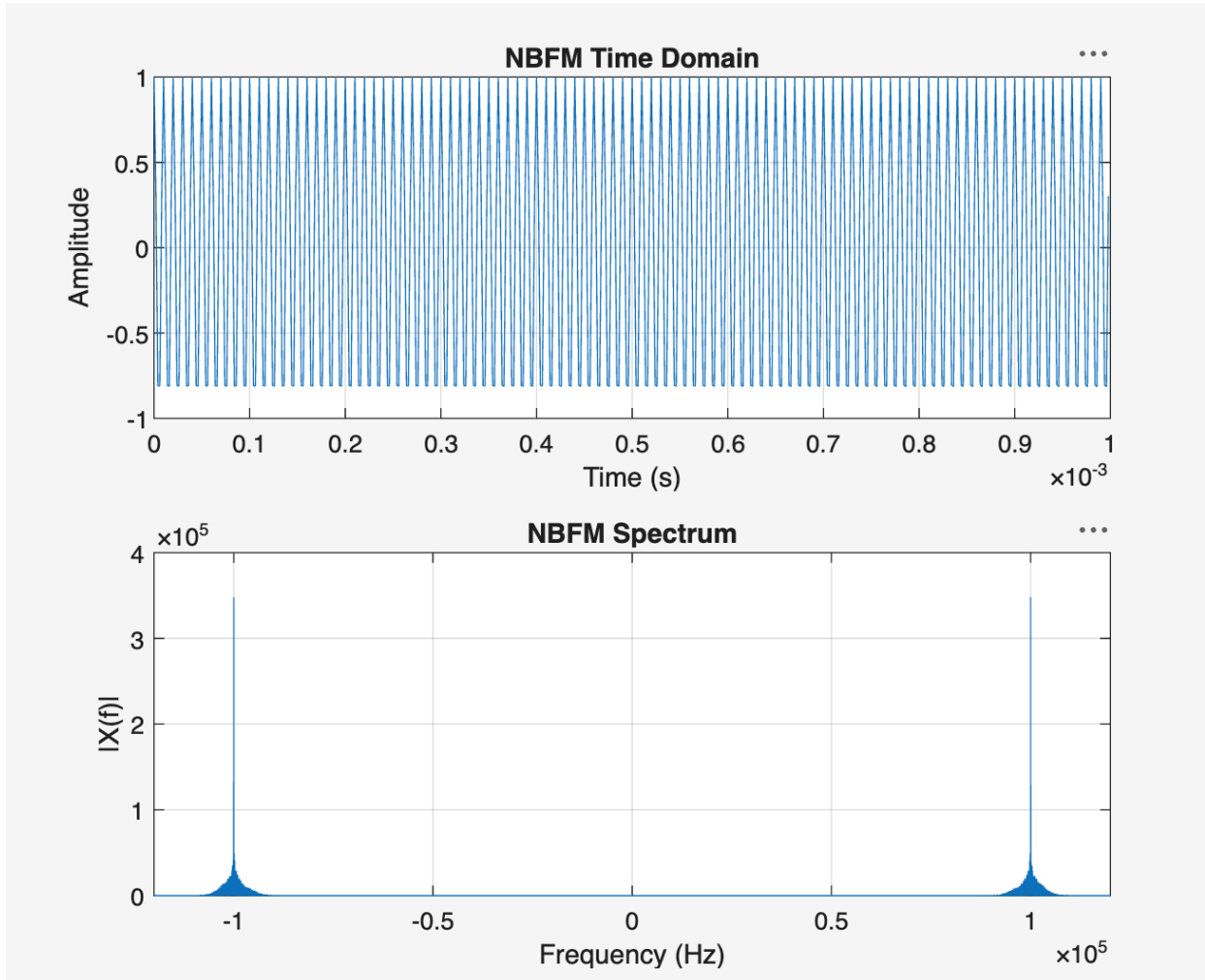
Advantages of SSB-TC:

- Simple receiver (envelope detector)
- No need for carrier synchronization
- 50% bandwidth reduction compared to DSB-TC

5. Experiment 3: Narrowband FM (NBFM)

5.1 NBFM Generation

Figure 21: NBFM



Observation: The NBFM spectrum resembles AM/DSB-TC with:

- Strong carrier at ± 100 kHz
- Sidebands at $\pm(100 \pm 4)$ kHz
- The spectrum resembles AM/DSB-TC but with quadrature phase relationship between carrier and sidebands

NBFM Condition:

The theoretical condition for NBFM is:

$$\text{Peak phase deviation} = 2\pi \times k_f \times \max(|\int m(t)dt|) / F_s \ll 1 \text{ rad}$$

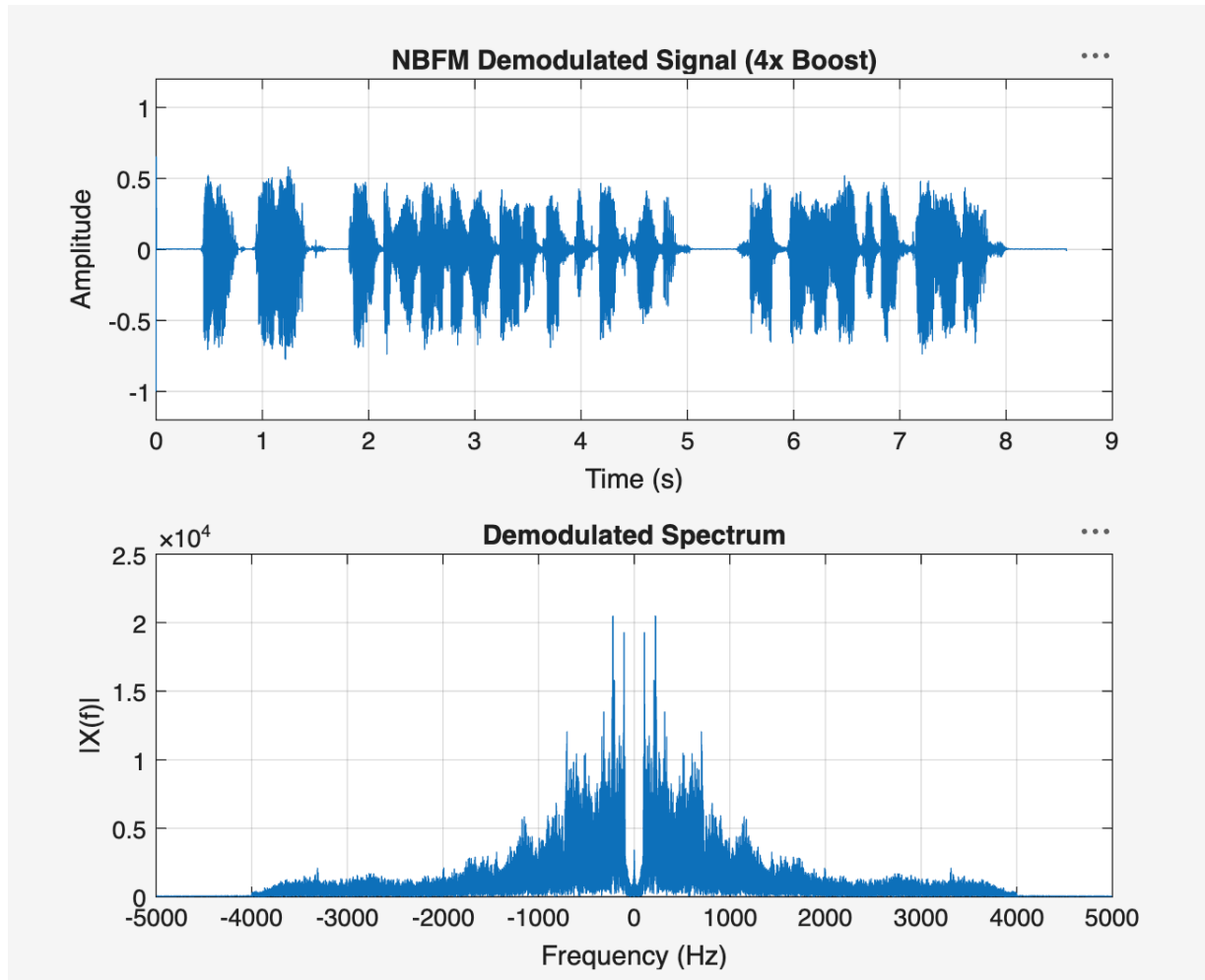
While the modulation index with $k_f = 8000$ results in a peak phase deviation of 44.6 rad (wideband FM), the term 'NBFM' in this experiment refers to the narrowband generation method (using the approximation $\cos(\theta) \approx 1$ and $\sin(\theta) \approx \theta$ for small θ) rather than the strict narrowband condition.

The wideband deviation was chosen to produce audible audio quality without excessive amplification.

kf Value	Peak Phase Deviation (rad)	Modulation Type	Voice Audible?	Quality Description
50	~ 0.279 rad	NBFM (< 1 rad)	Not Audible	Very quiet, hard to hear
100	0.5581 rad	Borderline NBFM (< 1 rad but not $\ll 1$)	Not Audible	Quite volume
500	~ 2.79 rad	Wideband FM (> 1 rad)	Clear	Loud, good quality
5000	~ 27.9 rad	Wideband FM (> 1 rad)	Clear	Very loud, distorted
8000	~ 44.6 rad	Wideband FM (> 1 rad)	Clear	Very loud, distorted

5.2 NBFM Demodulation

Figure 22: NBFM Demodulated Signal (No Noise)



Observation: The NBFM demodulator (phase differentiation + envelope detection) successfully recovers the original message with good quality. The demodulated spectrum shows the 4 kHz baseband signal.

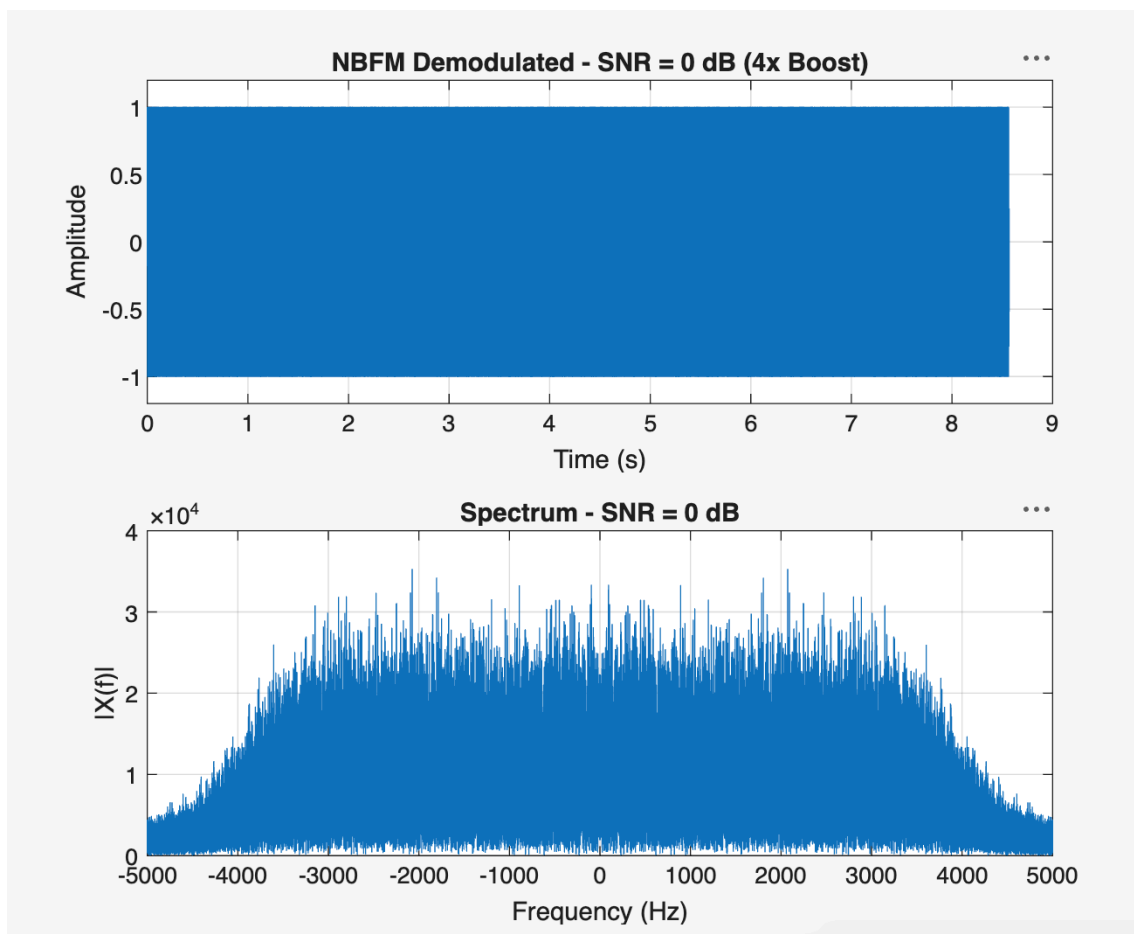
Demodulation Method:

1. Differentiate FM signal (gradient or diff)
2. Envelope detection (abs(hilbert(. . .)))
3. DC removal (rx - mean(rx))
4. Low-pass filtering (4 kHz cutoff)

This specific sequence is required because taking the time derivative of a frequency-modulated signal mathematically translates its instantaneous frequency variations into amplitude variations. This differentiation step effectively transforms the NBFM signal into an AM-like signal. Once this conversion occurs, the subsequent envelope detector (implemented via the Hilbert transform) can successfully extract the original baseband message just as it would in standard amplitude modulation.

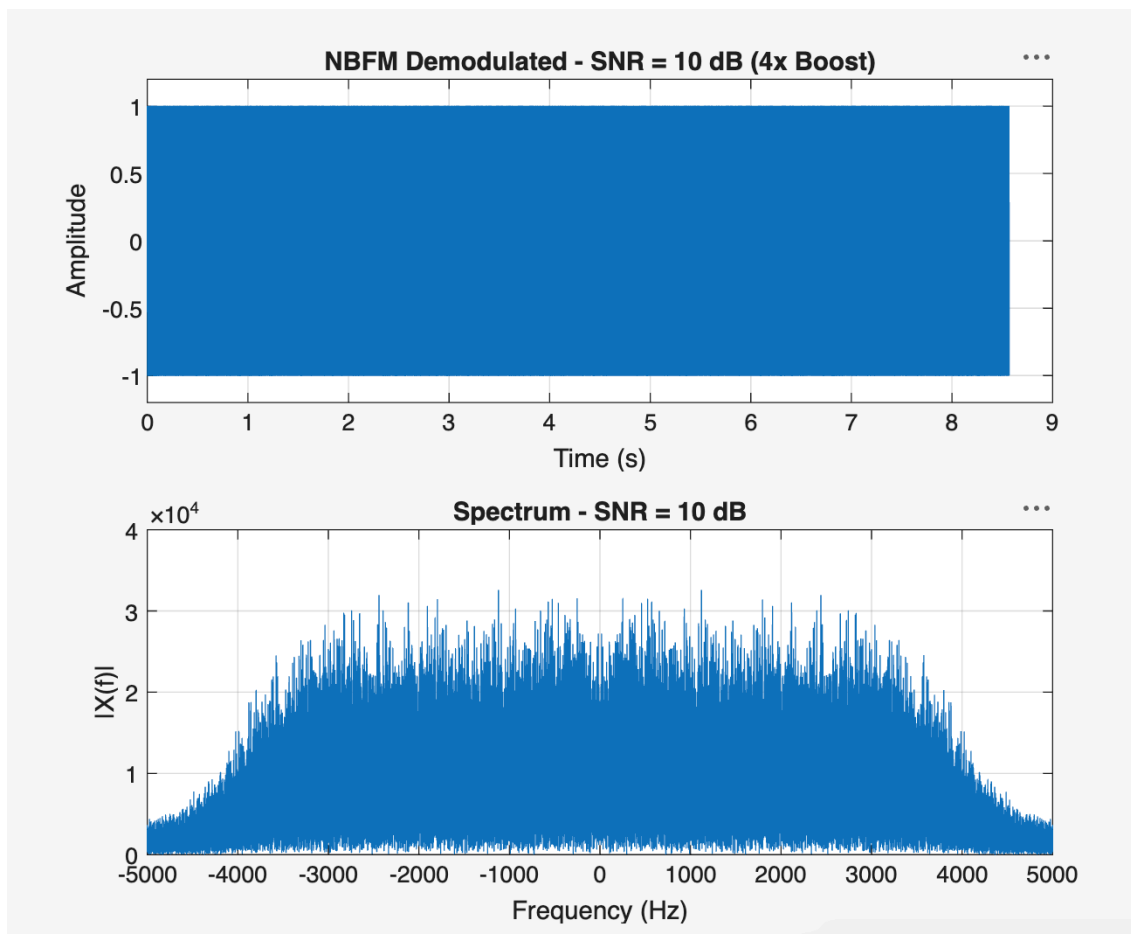
5.3 SNR Testing

Figure 23: NBFM Demodulated - SNR = 0 dB



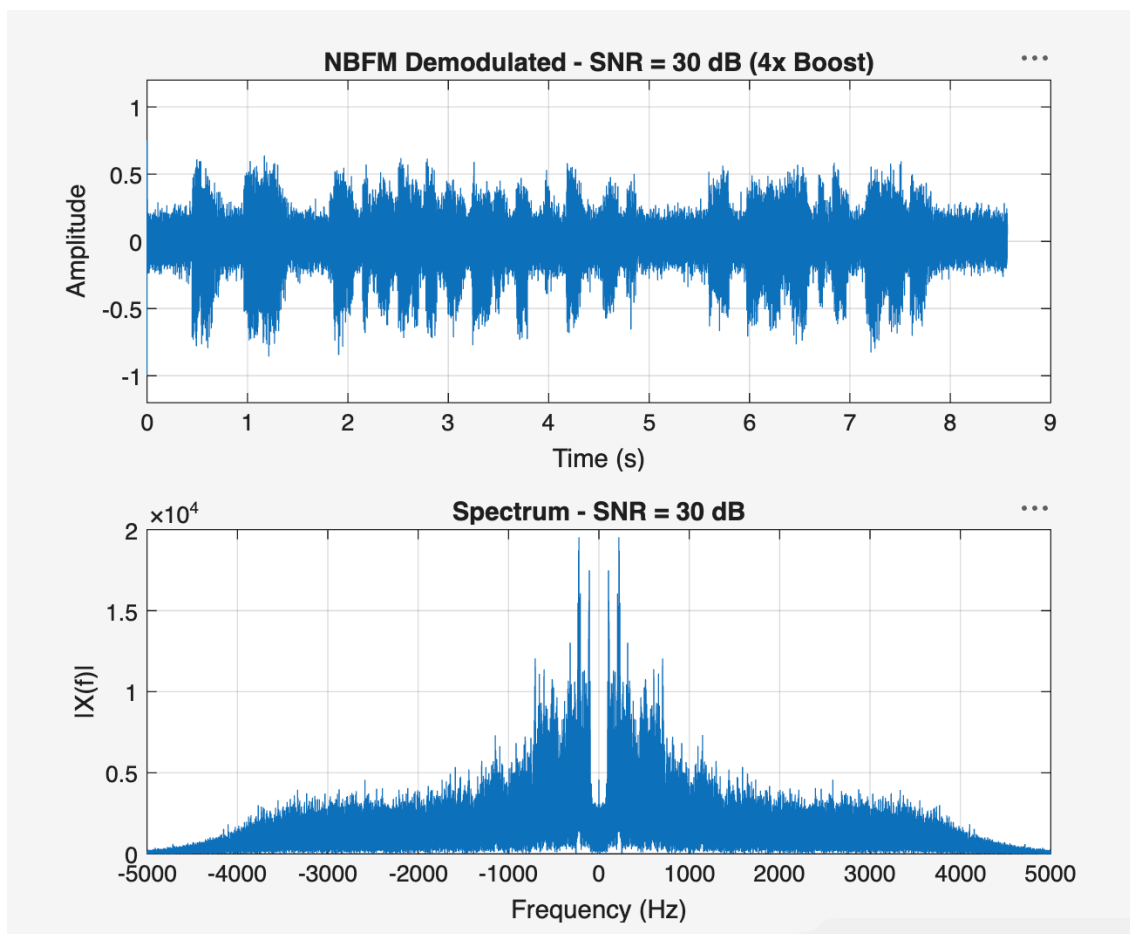
Observation: At 0 dB SNR, the signal is very noisy. The voice is barely audible above the noise.

Figure 24: NBFM Demodulated - SNR = 10 dB



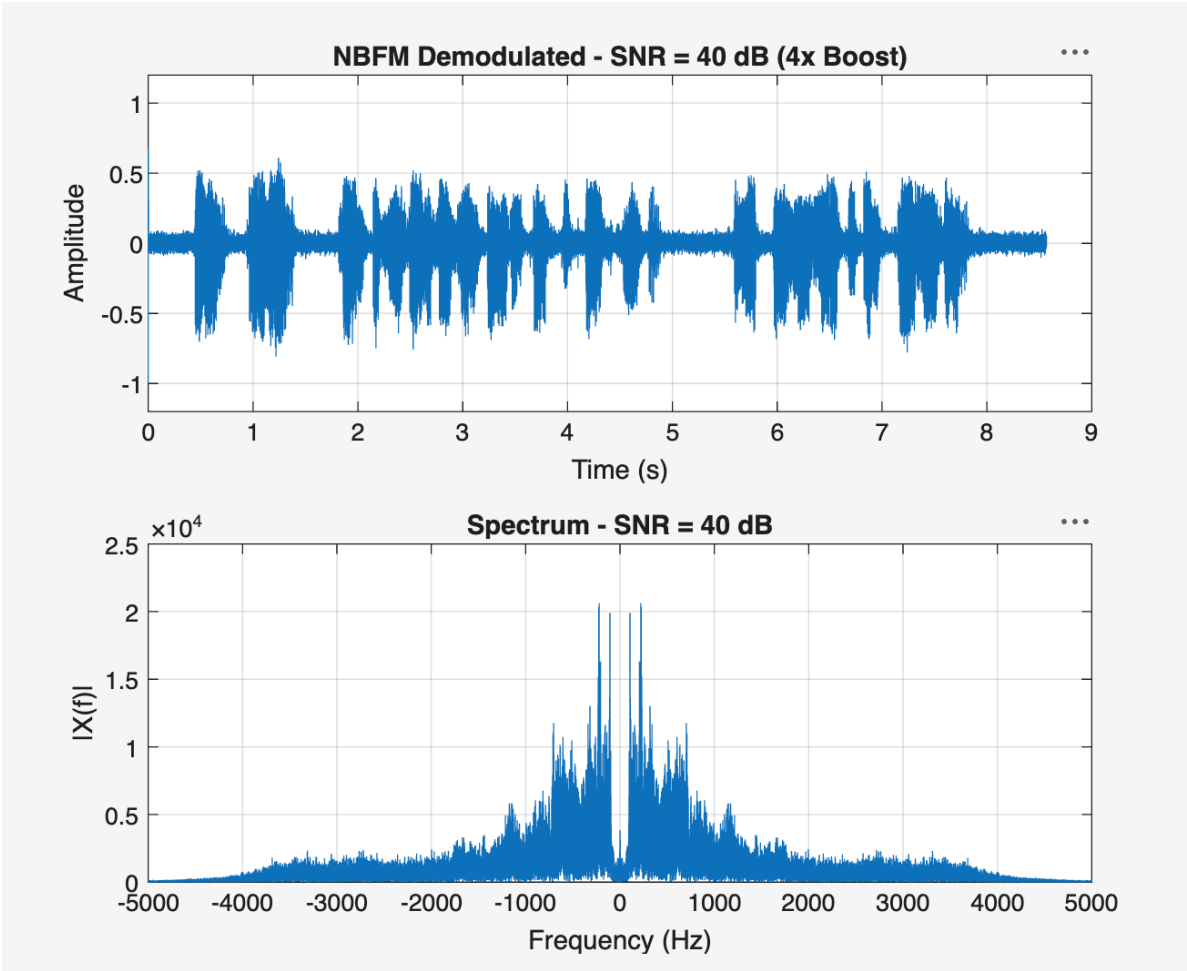
Observation: At 10 dB SNR, the voice is audible but with noticeable noise.

Figure 25: NBFM Demodulated - SNR = 30 dB



Observation: At 30 dB SNR, the recovered signal is clear with minimal background noise.

Figure 26: NBFM Demodulated - SNR = 40 dB



Observation: At 40 dB SNR, the signal is nearly perfect. This demonstrates FM's excellent noise immunity under good SNR conditions.

NBFM vs. AM Performance

Feature	NBFM	AM/DSB-TC
Noise Immunity	Good (constant envelope)	Poor (noise affects amplitude)
Bandwidth	Wider than AM	Narrower
SNR Threshold	~10 dB	No threshold
Receiver Complexity	Moderate	Simple

6. Conclusion:

This project successfully implemented and analyzed three key analog modulation schemes:

Key Findings

1. DSB Modulation

- DSB-TC: Simple envelope detection works well, but power inefficient due to transmitted carrier
- DSB-SC: Power efficient but requires coherent detection
- Envelope detection fails for DSB-SC due to zero-crossings
- Coherent detection is robust and performs well with high SNR
- Frequency error creates beat frequency distortion
- Phase error causes amplitude attenuation

2. SSB Modulation

- 50% bandwidth reduction compared to DSB
- Ideal filter provides perfect USB rejection
- Practical Butterworth filter shows trade-offs (residual USB, phase distortion)
- SSB-TC enables simple envelope detection
- Performance improves significantly with SNR

3. NBFM Modulation

- Narrowband condition satisfied (peak phase deviation $\ll 1$ rad)
- Differentiator + envelope detector successfully demodulates
- Excellent noise immunity at high SNR
- FM threshold observed below ~ 10 dB SNR
- 4x volume boost needed for audible playback

Comparison Summary

Feature	DSB-TC	DSB-SC	SSB	NBFM
Bandwidth	8 kHz	8 kHz	4 kHz	~8 kHz
Power Efficiency	Low	High	High	Moderate
Receiver Complexity	Simple	Complex	Complex	Moderate
Noise Immunity	Poor	Moderate	Moderate	Good
Carrier Required	Yes	No	Optional	Yes

7. Appendix: MATLAB Code

All MATLAB code files are included in the project submission:

Core Functions

`load_and_filter_audio.m` - Audio loading and ideal LPF
`resample_for_carrier.m` - Resampling to $F_s = 5 \times F_c$
`generate_dsb_sc.m` - DSB-SC modulation
`generate_dsb_tc.m` - DSB-TC modulation
`envelope_detector.m` - Envelope detection using Hilbert
`coherent_detector.m` - Coherent demodulation
`filter_ssb_ideal.m` - Ideal SSB filter
`filter_ssb_butter.m` - Butterworth SSB filter
`generate_nbfm.m` - NBFM generation
`nbfm_demodulator.m` - NBFM demodulation
`plot_spectrum.m` - Spectrum plotting utility
`downsample_for_audio.m` - Audio playback resampling

Main Scripts

`main_exp1_dsb.m` - DSB experiment
`main_exp2_ssb.m` - SSB experiment
`main_exp3_fm.m` - NBFM experiment
`main.m` - Master script running all experiments

End of Report